

Universitat Politècnica de Catalunya

ESEIAAT

Applied Subsystem Design

Attitude Determination Control Systems

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1 GEOS18 Controlled Rotation

The geostationary satellite GOES 18 is required to rotate 45º around its Z axis, otherwise expressed:

$$\alpha = 45^\circ \qquad \vec{e} = [1, 0, 0] \tag{1}$$

From this form, and using expressions [XXXX] we can obtain:

$$\vec{q} = 0.9239 + 0.3827\vec{k} \tag{2}$$

$$DCM = \begin{bmatrix} 0.7071 & -0.7071 & 0 \\ 0.7071 & 0.7071 & 0 \\ 0 & 0 & 1.0000 \end{bmatrix} \tag{3}$$

The validity of these rotations can be assessed through checking that:

$$||\vec{q}|| = 1 \qquad DCM^{-1} \cdot DCM = I$$

2 NOAA21 Reaction Wheel Rotation

NOAA-21 spaceship has a reaction wheel torque that produces a single rotation of 120º around [1,1,1], therefore:

$$\alpha = 120^\circ \qquad \vec{e} = \frac{1}{\sqrt{3}}[1, 1, 1] \tag{4}$$

Applying the same operations as in the previous section, we can turn this rotation to quaternion and DCM representation:

$$\vec{q} = 0.5 \cdot (1 + \vec{i} + \vec{j} + \vec{k}) \tag{5}$$

$$DCM = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \tag{6}$$

3 Apollo 31 particular rotation

Apollo mission's DCS reads the following spaceship attitude:

$$[\phi, \theta, \psi] = [30, 90, 45]^\circ \tag{7}$$

This rotation can be re-expressed in quaternion and DCM notation as:

$$\vec{q} = 0.7011 - 0.0923\vec{i} + 0.7011\vec{j} - 0.0923\vec{k} \tag{8}$$

$$DCM = \begin{bmatrix} 0 & -0.2588 & 0.9659 \\ 0 & 0.9659 & 0.2588 \\ -1 & 0 & 0 \end{bmatrix} \tag{9}$$

If we try to obtain the Euler axis from the previous DCM, we obtain the following results:

$$[\phi, \theta, \psi] = [165, 90, 0]^\circ \tag{10}$$

As it can be seen, these angles are not the same as the original ones. This is due to a gimbal lock occuring, as the central angle of the ZYX system (θ) is exactly 90°.

This is the perfect example on why Euler angles are not a roubust enough representation for ADCS, being susceptible to singularities that do not appear when using other approaches such as quaternions.

4 Cerisse Reveal Move

After being hit by a space debris, the Frenc secret misison Cerisse was uncovered and rotated 180° around its x axis, otherwise expressed as:

$$\alpha = 180^\circ \qquad \vec{e} = [1, 0, 0] \tag{11}$$

And through the same algorithms, we can also express this rotation as:

$$\vec{q} = \vec{i} \tag{12}$$

$$DCM = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \tag{13}$$

If a simple direct algorithm is run in order to obtain euler angles back from this DCM, the following result is obtained:

$$[\psi, \theta, \phi] = [180, 0, 0]$$

5 Star Tracker Controlled Rotation

For the star tracked to be oriented to a certain objective, it needs to rotate 60° around its x axis, what can be expressed as:

$$\alpha = 60^{\circ} \qquad \vec{e} = [1, 0, 0] \tag{14}$$

Applying the same operations as in the previous section, we can turn this rotation to quaternion notation:

$$\vec{q} = 0.1543 - 0.9880\vec{i} \tag{15}$$

This rotation is also valid, as it fulfills:

$$\vec{q}^{-1} \cdot \vec{q} = 1 \quad \text{and} \quad ||\vec{q}|| = 1$$

6 Conclusions and observations

Through this deliverable, several cases of gimbal lock have been found, cases in which quaternions have proved not to show any kind of numerical singularity.

Independently, in the theory class documents, a transformation matrix was transposed, what lead to us, the students, to find non-coherent results, such as the transformation

$$45^{\circ}, [0, 0, 1] \xrightarrow{to_DCM} DCM \xrightarrow{to_angle_axis} 45^{\circ}, [0, 0, -1] \tag{16}$$

This error was found during class session, assessed with the professor, and solved in a matter of minutes, therefore ensuring the quality and robustness of the developed algorithms.